WEEK 1 EXECUTION PLAN

Fintech Backend Engineering Roadmap • Month 1 Foundations

FOCUS AREA	DSA GOAL	DAILY COMMITMENT
Core Java & OOPs Pillars	**12 Array & String Problems**	**3-4 Hours Daily Practice**

7-Day Execution Schedule

DAY	JAVA OOPS PRACTICE & MINI-PROJECT	DSA PROBLEMS (LEETCODE DIRECT LINKS)
Day 1	**Classes, Objects & Encapsulation** Create `BankAccount.java` with private fields & public getters/setters.	1. Contains Duplicate 2. Majority Element
Day 2	**Inheritance & `super` Keyword** Extend `Account.java` into `SavingsAccount` & `CheckingAccount`.	3. Valid Anagram 4. Missing Number
Day 3	**Polymorphism (Overload/Override)** Practice method overloading & overriding in `Transaction.java`.	5. Two Sum 6. Merge Sorted Array
Day 4	**Abstraction & Interfaces** Build `PaymentGateway` interface & implement `UpiPayment.java`.	7. Best Time to Buy & Sell Stock 8. Intersection of Two Arrays II
Day 5	**Mini Console Bank System** Combine all OOPs classes into a functional `BankSystem.java` runner.	9. Move Zeroes 10. Remove Duplicates from Sorted Array
Day 6	**Access Modifiers & `static` / `final`** Add auto-increment account numbers & interest constants.	11. Reverse String 12. Valid Palindrome
Day 7	**Code Cleanup & GitHub Review** Refactor Java files, format code, and update `README.md` ticks.	13. Find Pivot Index (Bonus) • Complete Revision of All Solved Problems

Week 1 GitHub Directory Target

```
java-backend-roadmap/ └── month-01-java-dsa-foundations/ ├── week-1-oops/ | ├── BankAccount.java | ├── SavingsAccount.java | ├── PaymentGateway.java | └── BankSystem.java └── dsa-40-easy/ ├── TwoSum.java ├── ValidAnagram.java └── ... (12 Solved LeetCode Files)
```

Daily Workflow Checklist

- [ ] Write clean Java code following OOP rules
- [ ] Solve & test 2 LeetCode problems daily
- [ ] Save files in `month-01-java-dsa-foundations`
- [ ] Run `git add .` & `git push` daily